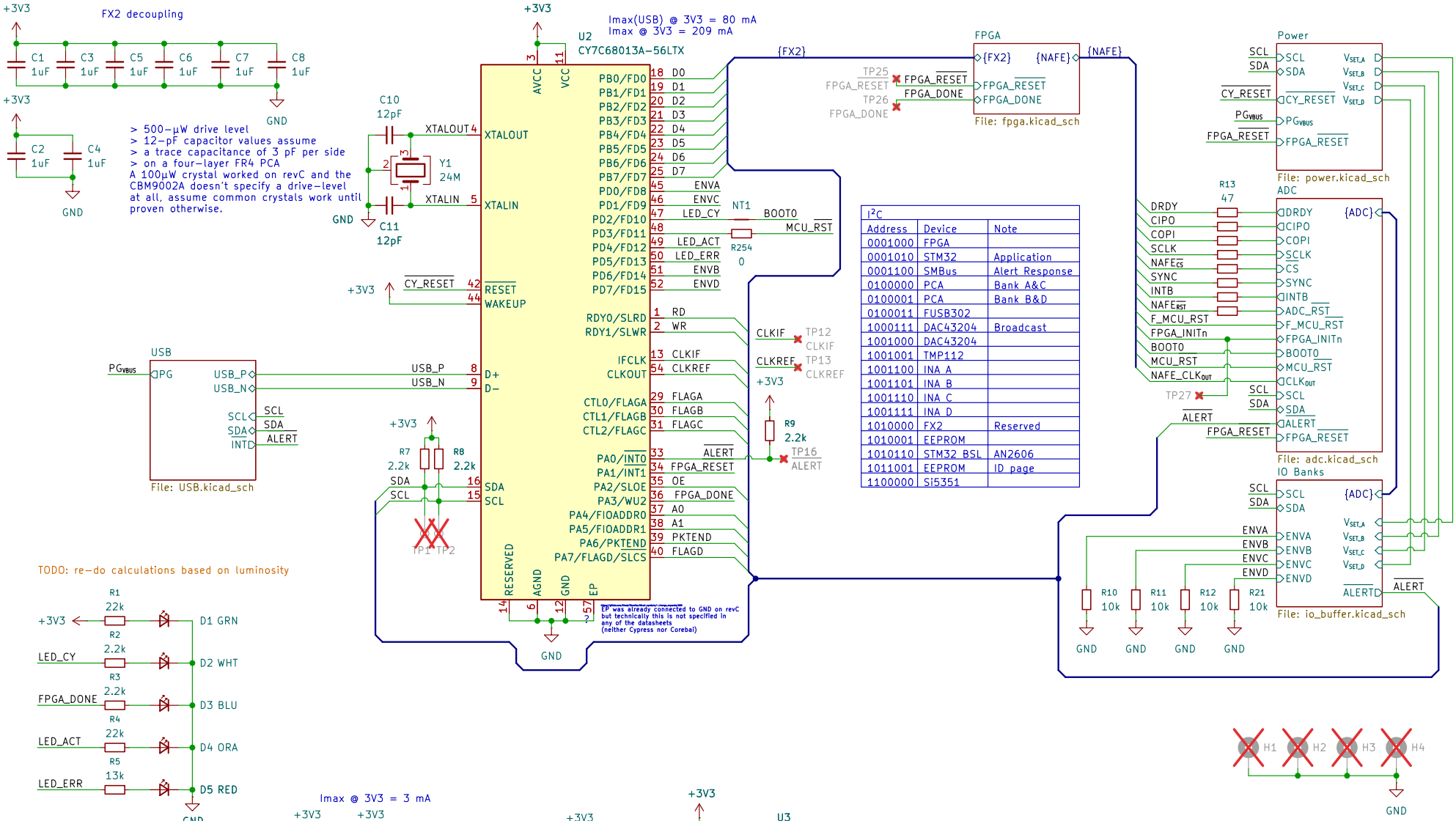
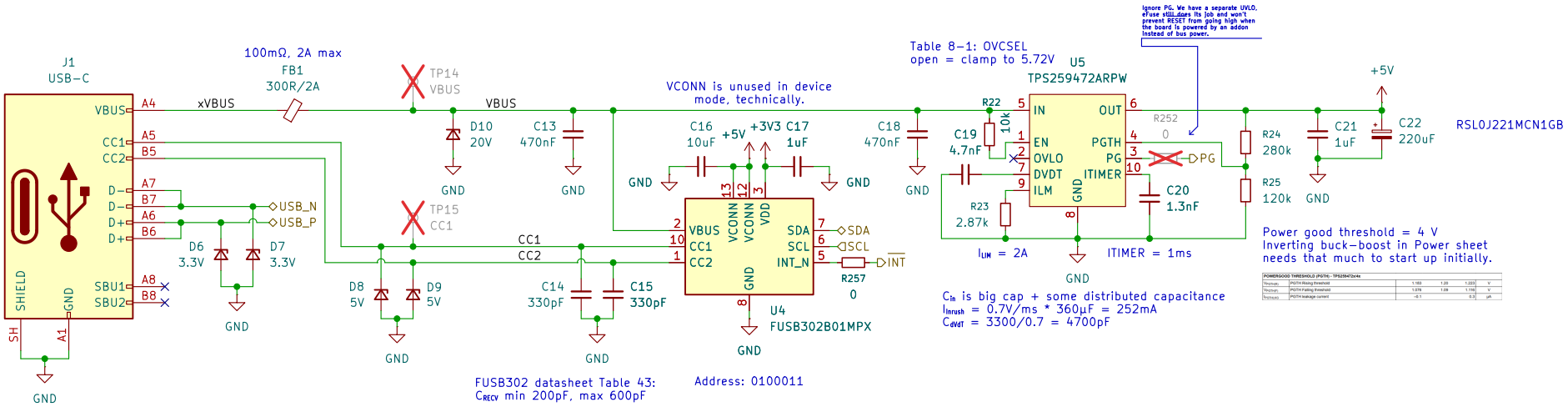

FX2 decoupling
C1 1uF
C3 1uF
C5 1uF
C6 1uF
C7 1uF
C8 1uF
C2 1uF
C4 1uF
> 500−µW drive level
> 12−pF capacitor values assume
> a trace capacitance of 3 pF per side
> on a four−layer FR4 PCA
A 100µW crystal worked on revC and the CBM9002A doesn't specify a drive−level at all, assume common crystals work until proven otherwise.
C10 12pF
C11 12pF
Y1 24M
XTALOUT
XTALIN
Imax(USB) @ 3V3 = 80 mA
Imax @ 3V3 = 209 mA
U2
CY7C68013A−56LTX
AVCC
VCC
RESET
WAKEUP
CY_RESET
PB0/FD0 D0
PB1/FD1 D1
PB2/FD2 D2
PB3/FD3 D3
PB4/FD4 D4
PB5/FD5 D5
PB6/FD6 D6
PB7/FD7 D7
PD0/FD8 ENVA
PD1/FD9 ENVC
PD2/FD10 LED_CY
PD3/FD11 LED_ACT
PD4/FD12 LED_ERR
PD5/FD13 ENVB
PD6/FD14 ENVD
PD7/FD15
RDY0/SLRD RD
RDY1/SLWR WR
IFCLK CLKIF
CLKOUT CLKREF
CTL0/FLAGA FLAGA
CTL1/FLAGB FLAGB
CTL2/FLAGC FLAGC
PA0/INT0 ALERT
PA1/INT1 FPGA_RESET
PA2/SLOE OE
PA3/WU2 FPGA_DONE
PA4/FIOADDR0 A0
PA5/FIOADDR1 A1
PA6/PKTEND PKTEND
PA7/FLAGD/SLCS FLAGD
RESERVED
AGND
GND
EP
EP was already connected to GND on revC but technically this is not specified in any of the datasheets (neither Cypress nor Corebal)
NT1
BOOT0
MCU_RST
R254 0
R7 2.2k
R8 2.2k
R9 2.2k
TP1
TP2
TP16 ALERT
TP12 CLKIF
TP13 CLKREF
USB
PG_VBUS
USB_P
USB_N
D+
D−
SCL
SDA
INT
ALERT
File: USB.kicad_sch
{FX2}
FPGA
{NAFE}
TP25 FPGA_RESET
TP26 FPGA_DONE
File: fpga.kicad_sch
Power
CY_RESET
PG_VBUS
FPGA_RESET
File: power.kicad_sch
ADC
R13 47
DRDY
CIPO
COPI
SCLK
NAFE_CS
SYNC
INTB
NAFE_RST
F_MCU_RST
FPGA_INITn
BOOT0
MCU_RST
NAFE_CLK_OUT
TP27
ADC_RST
CLK_OUT
File: adc.kicad_sch
IO Banks
{ADC}
VSET_A
VSET_B
VSET_C
VSET_D
ENVA
ENVB
ENVC
ENVD
R10 10k
R11 10k
R12 10k
R21 10k
File: io_buffer.kicad_sch
I²C
Address | Device | Note
0001000 | FPGA
0001010 | STM32 | Application
0001100 | SMBus | Alert Response
0100000 | PCA | Bank A&C
0100001 | PCA | Bank B&D
0100011 | FUSB302
1000111 | DAC43204 | Broadcast
1001000 | DAC43204
1001001 | TMP112
1001100 | INA A
1001101 | INA B
1001110 | INA C
1001111 | INA D
1010000 | FX2 | Reserved
1010001 | EEPROM
1010110 | STM32 BSL | AN2606
1011001 | EEPROM | ID page
1100000 | Si5351
TODO: re−do calculations based on luminosity
R1 22k D1 GRN
R2 2.2k D2 WHT
R3 2.2k D3 BLU
R4 22k D4 ORA
R5 13k D5 RED
LED_CY
FPGA_DONE
LED_ACT
LED_ERR
Imax @ 3V3 = 3 mA
U1
FX2_MEM
BL24C256A−SFRC
R6 DNP
C9 1uF
RECOVER
Addr: 1010001
Release reset while shorting these pads if it ever stops enumerating
U3
TMP112xxDRL
C12 1uF
ADD0
Addr: 1001000
H1
H2
H3
H4
automated.ee
Glasgow Debug Tool
In / Fully Automated
Sheet: /
File: glasgow.kicad_sch
Title: FX2 & Power Input
Size: A4
Date: 2026−09−22
Rev: D1
KiCad E.D.A. 10.0.6
Id: 1/11

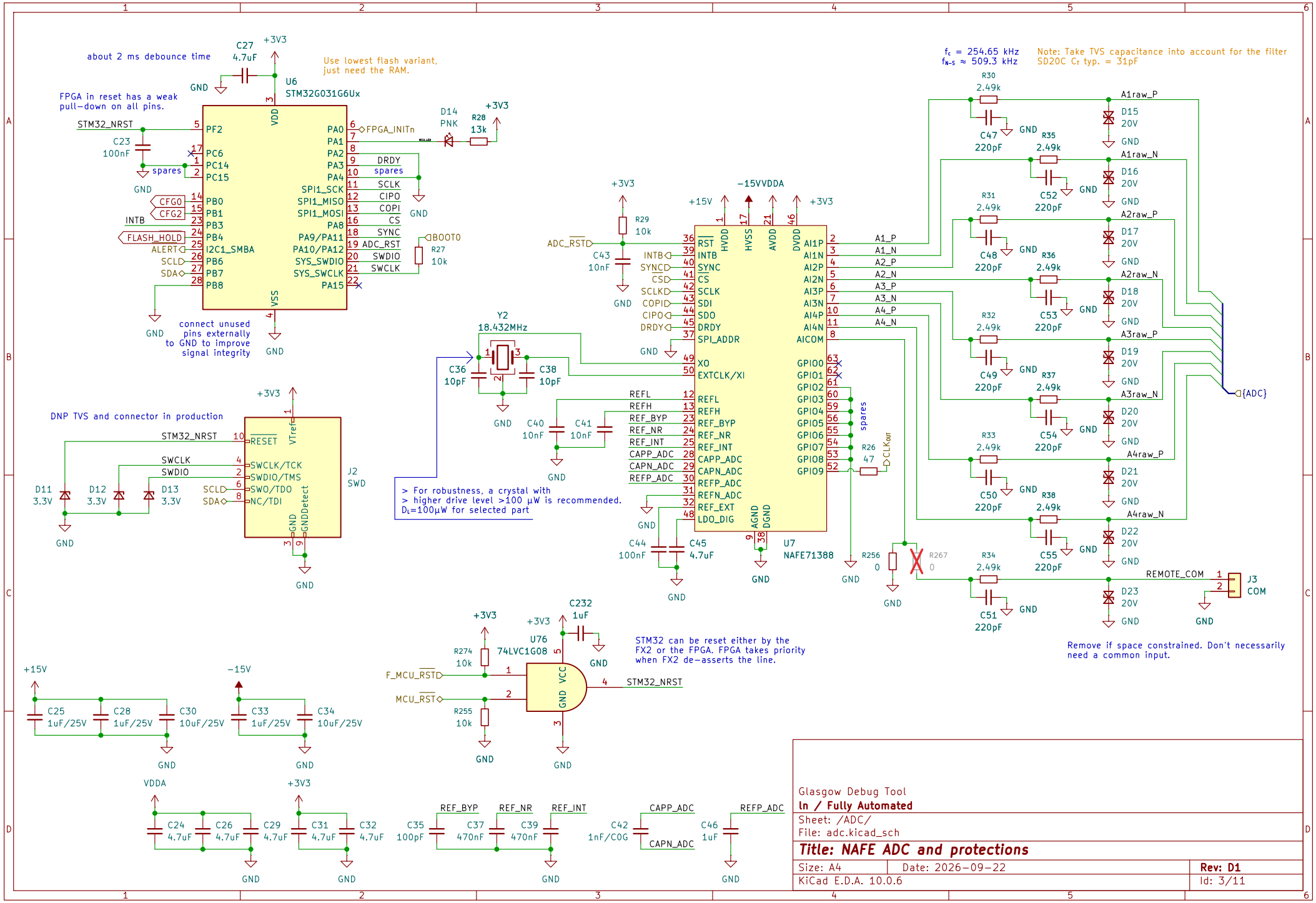


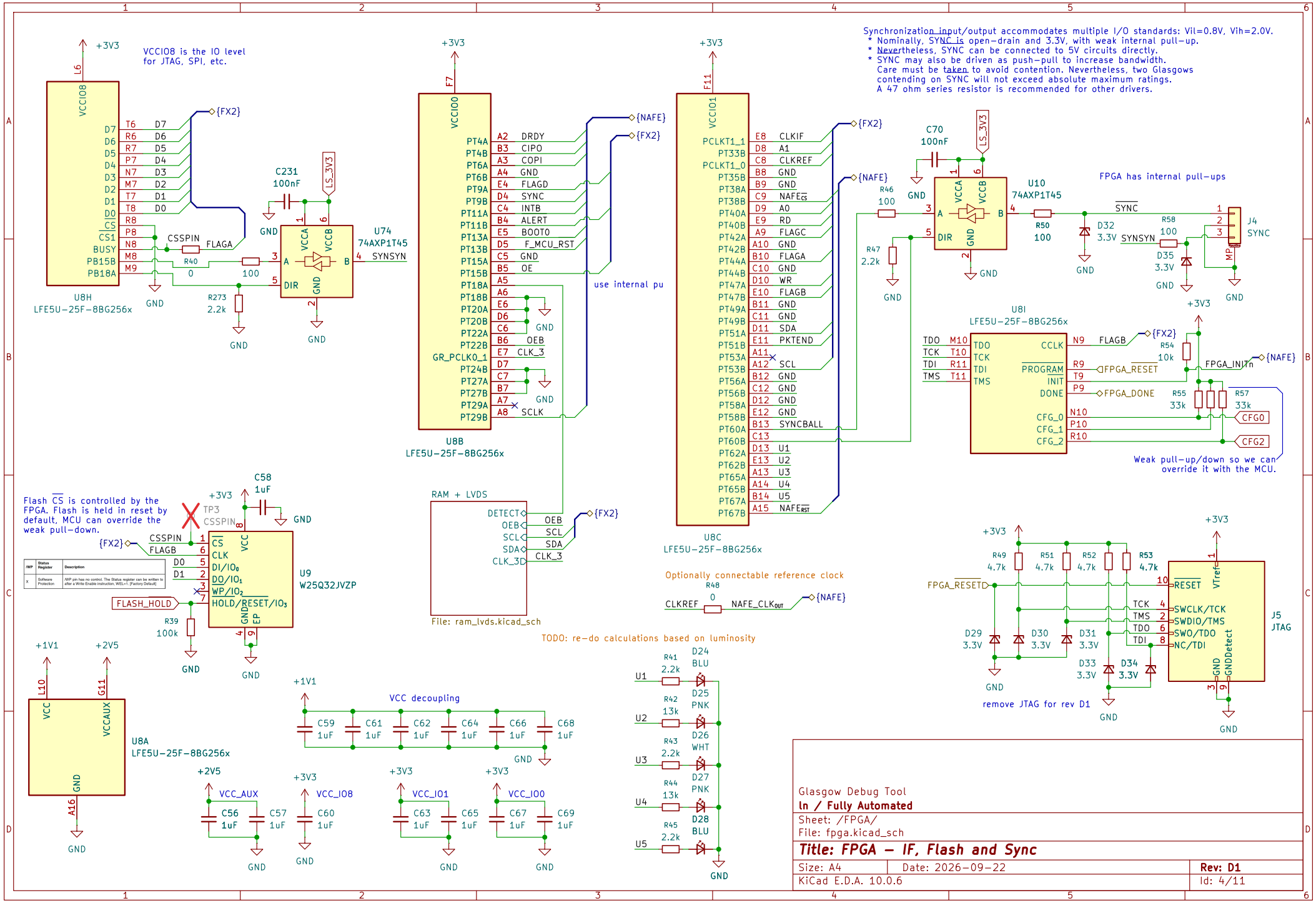
Glasgow Debug Tool
In / Fully Automated

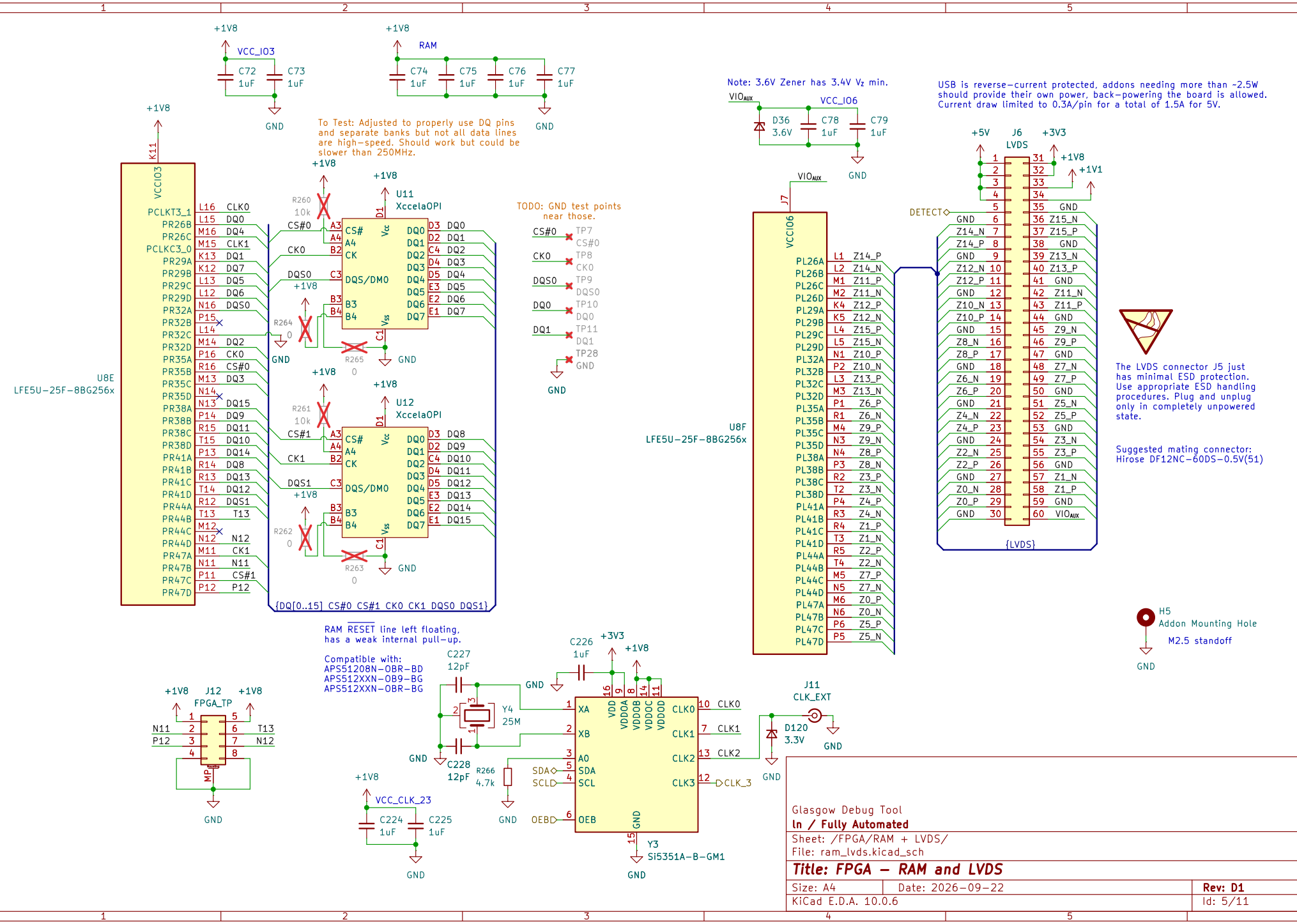
Sheet: /USB/
File: USB.kicad_sch

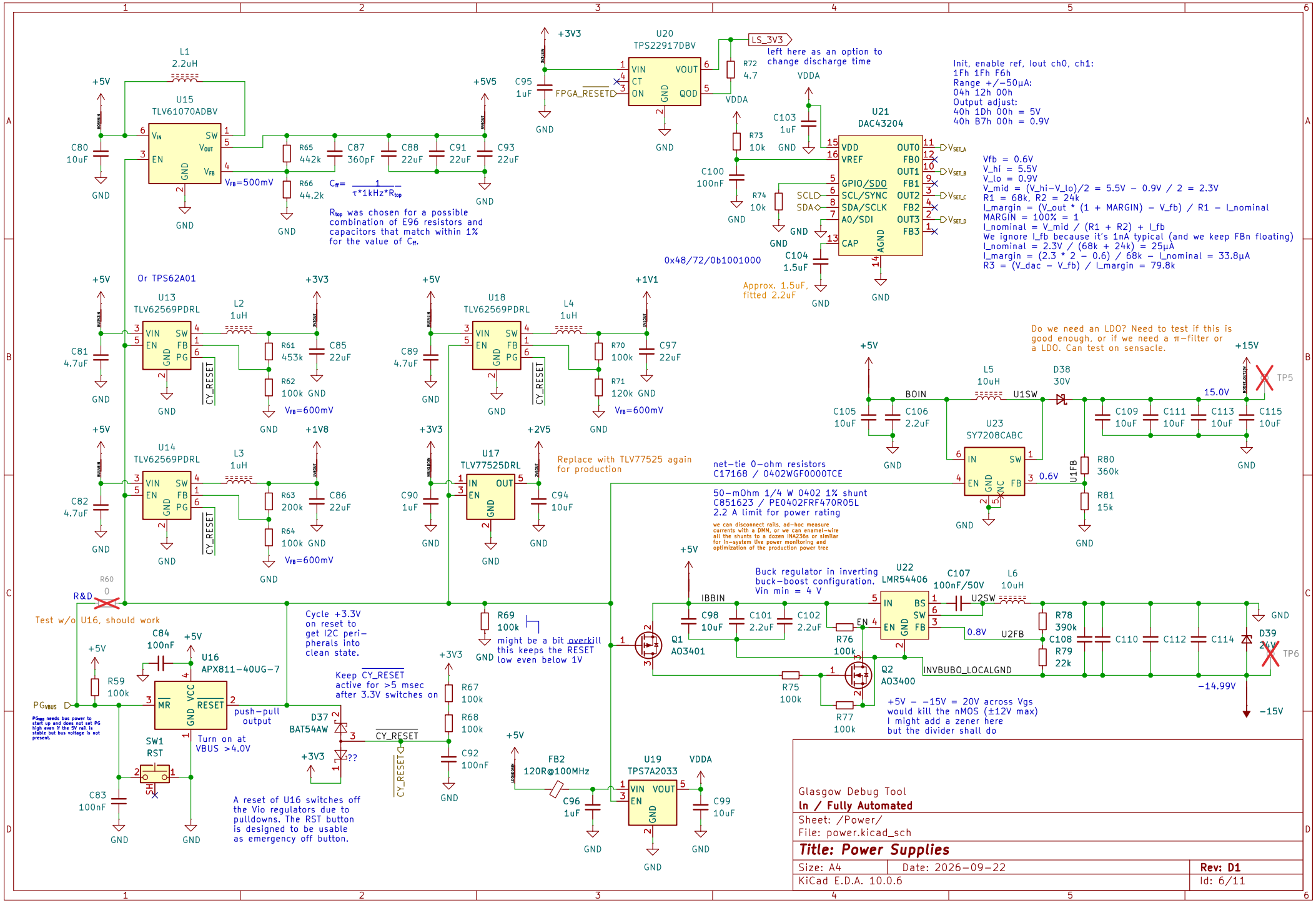
Title: USB input protection

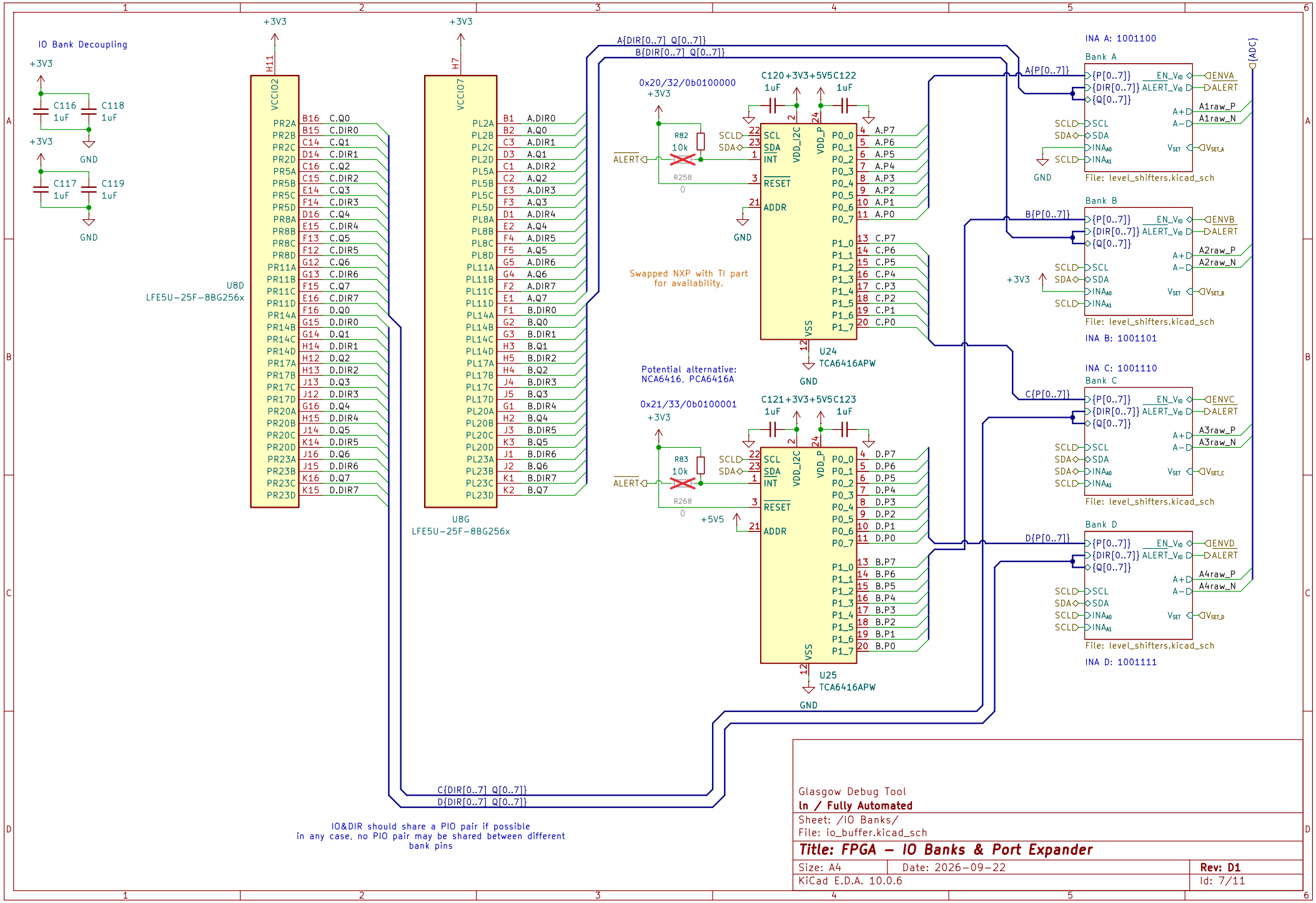
Size: A4	Date: 2026−09−22	**Rev: D1**
KiCad E.D.A. 10.0.6		Id: 2/11

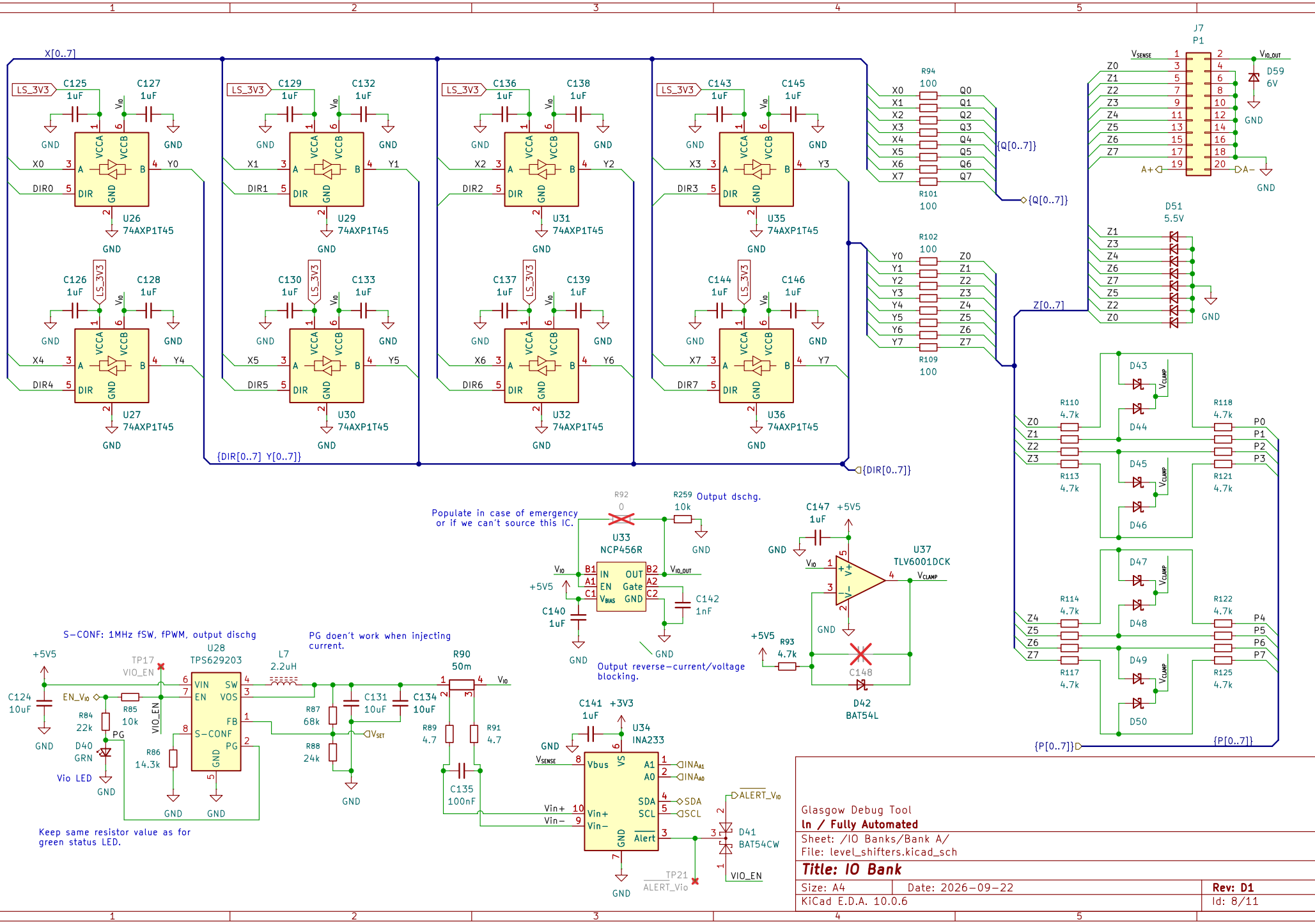

about 2 ms debounce time
C27
4.7uF
+3V3
GND
U6
STM32G031G6Ux
Use lowest flash variant, just need the RAM.
FPGA in reset has a weak pull-down on all pins.
STM32_NRST
C23
100nF
spares
INTB
CFG0
CFG2
FLASH_HOLD
ALERT
SCLD
SDA
PF2
PC6
PC14
PC15
PB0
PB1
PB3
PB4
I2C1_SMBA
PB6
PB7
PB8
VDD
VSS
PA0
PA1
PA2
PA3
PA4
SPI1_SCK
SPI1_MISO
SPI1_MOSI
PA8
PA9/PA11
PA10/PA12
SYS_SWDIO
SYS_SWCLK
PA15
FPGA_INITn
DRDY
spares
SCLK
CIPO
COPI
CS
SYNC
ADC_RST
SWDIO
SWCLK
BOOT0
R27
10k
D14
PNK
R28
13k
connect unused pins externally to GND to improve signal integrity
DNP TVS and connector in production
D11
3.3V
D12
3.3V
D13
3.3V
J2
SWD
VTref
RESET
SWCLK/TCK
SWDIO/TMS
SWO/TDO
NC/TDI
GNDDetect
Y2
18.432MHz
C36
10pF
C38
10pF
C40
10pF
C41
10pF
> For robustness, a crystal with
> higher drive level >100 µW is recommended.
Dₗ=100µW for selected part
R29
10k
C43
10nF
−15VVDDA
+15V
HVDD
HVSS
AVDD
DVDD
REFL
REFH
REF_BYP
REF_NR
REF_INT
CAPP_ADC
CAPN_ADC
REFP_ADC
REFN_ADC
REF_EXT
LDO_DIG
AGND
DGND
EXTCLK/XI
AI1P
AI1N
AI2P
AI2N
AI3P
AI3N
AI4P
AI4N
AICOM
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
U7
NAFE71388
C44
100nF
C45
4.7uF
R26
47
CLKout
R256
0
R267
0
f_c = 254.65 kHz
f_{N-S} ≈ 509.3 kHz
Note: Take TVS capacitance into account for the filter
SD20C C_T typ. = 31pF
R30
2.49k
C47
220pF
R35
2.49k
C52
220pF
R31
2.49k
C48
220pF
R36
2.49k
C53
220pF
R32
2.49k
C49
220pF
R37
2.49k
C54
220pF
R33
2.49k
C50
220pF
R38
2.49k
C55
220pF
R34
2.49k
C51
220pF
A1raw_P
A1raw_N
A2raw_P
A2raw_N
A3raw_P
A3raw_N
A4raw_P
A4raw_N
REMOTE_COM
D15
20V
D16
20V
D17
20V
D18
20V
D19
20V
D20
20V
D21
20V
D22
20V
D23
20V
J3
COM
{ADC}
Remove if space constrained. Don't necessarily need a common input.
C232
1uF
U76
74LVC1G08
R274
10k
R255
10k
F_MCU_RST
MCU_RST
STM32_NRST
STM32 can be reset either by the FX2 or the FPGA. FPGA takes priority when FX2 de-asserts the line.
C25
1uF/25V
C28
1uF/25V
C30
10uF/25V
C33
1uF/25V
C34
10uF/25V
C24
4.7uF
C26
4.7uF
C29
4.7uF
C31
4.7uF
C32
4.7uF
C35
100pF
C37
470nF
C39
470nF
C42
1nF/C0G
C46
1uF
REF_BYP
REF_NR
REF_INT
CAPP_ADC
CAPN_ADC
REFP_ADC
Glasgow Debug Tool
In / Fully Automated
Sheet: /ADC/
File: adc.kicad_sch
Title: NAFE ADC and protections
Size: A4
Date: 2026-09-22
Rev: D1
KiCad E.D.A. 10.0.6
Id: 3/11

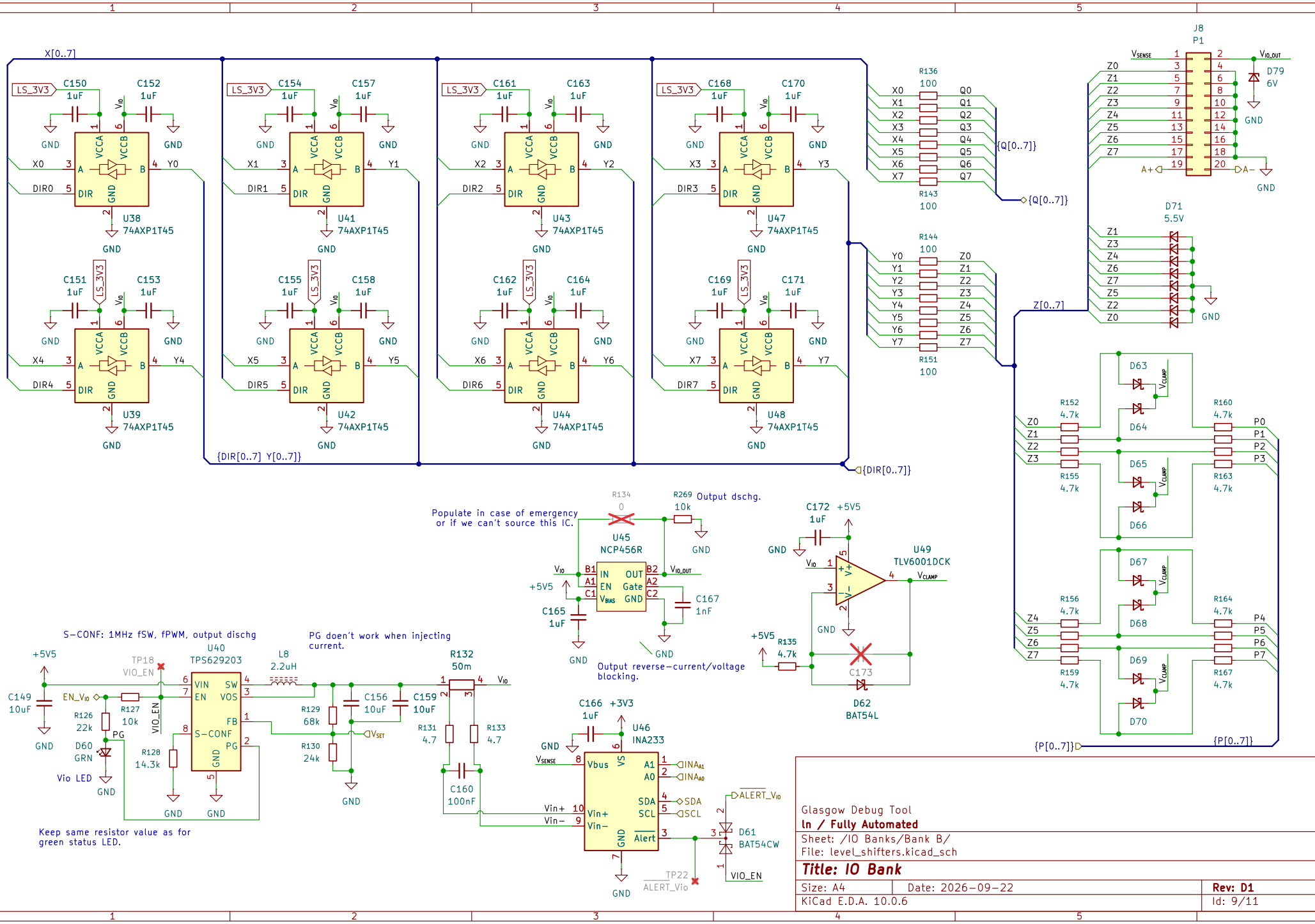

VCCIO8 is the IO level for JTAG, SPI, etc.
Synchronization input/output accommodates multiple I/O standards: Vil=0.8V, Vih=2.0V.
* Nominally, SYNC is open-drain and 3.3V, with weak internal pull-up.
* Nevertheless, SYNC can be connected to 5V circuits directly.
* SYNC may also be driven as push-pull to increase bandwidth.
Care must be taken to avoid contention. Nevertheless, two Glasgows contending on SYNC will not exceed absolute maximum ratings.
A 47 ohm series resistor is recommended for other drivers.
FPGA has internal pull-ups
Flash CS is controlled by the FPGA. Flash is held in reset by default, MCU can override the weak pull-down.
Weak pull-up/down so we can override it with the MCU.
use internal pu
RAM + LVDS
File: ram_lvds.kicad_sch
Optionally connectable reference clock
TODO: re-do calculations based on luminosity
VCC decoupling
remove JTAG for rev D1
U8H LFE5U-25F-8BG256x
U8B LFE5U-25F-8BG256x
U8C LFE5U-25F-8BG256x
U8I LFE5U-25F-8BG256x
U8A LFE5U-25F-8BG256x
U9 W25Q32JVZP
U10 74AXP1T45
U74 74AXP1T45
J4 SYNC
J5 JTAG
Glasgow Debug Tool
In / Fully Automated
Sheet: /FPGA/
File: fpga.kicad_sch
Title: FPGA - IF, Flash and Sync
Size: A4
Date: 2026-09-22
Rev: D1
KiCad E.D.A. 10.0.6
Id: 4/11

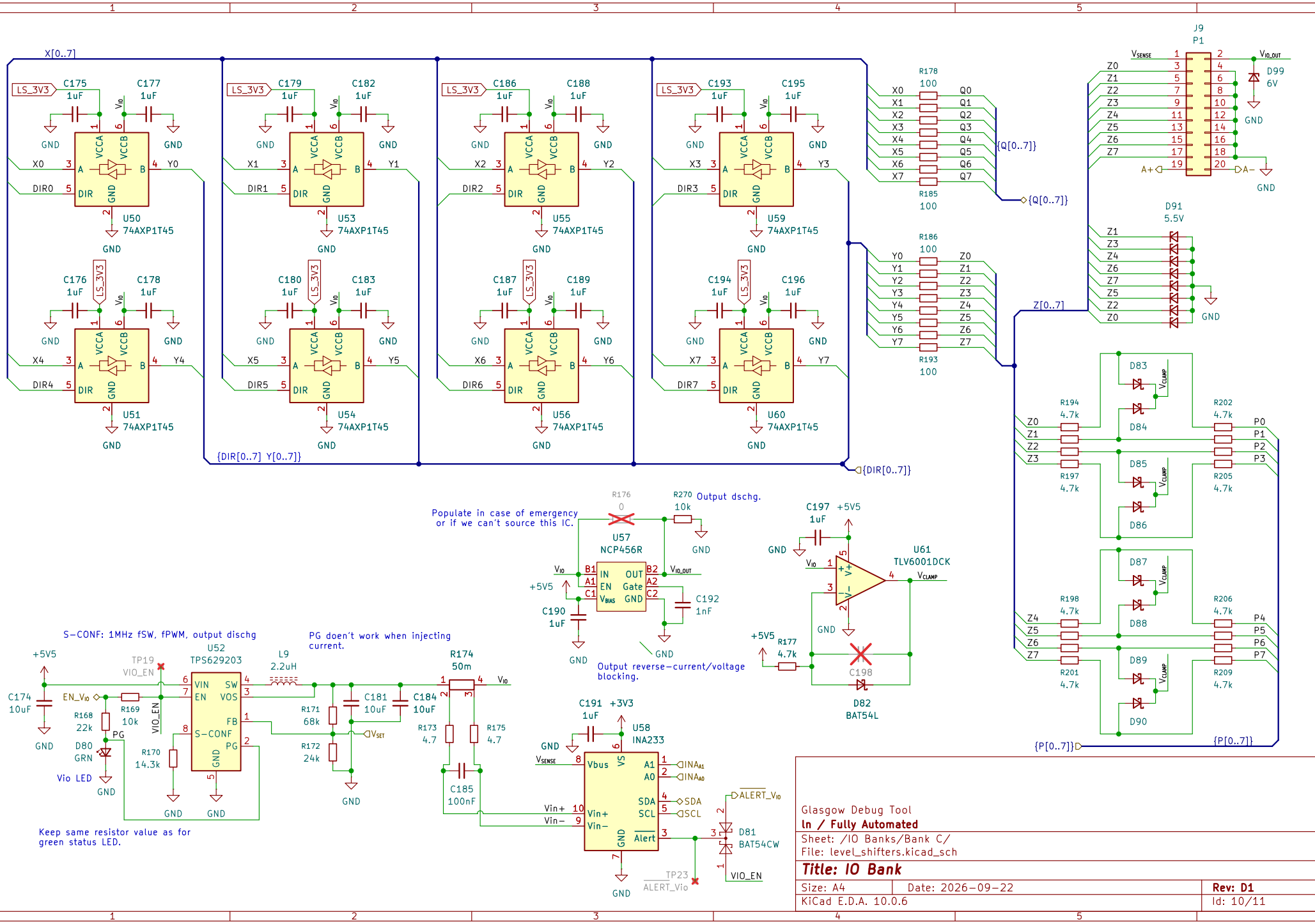

+1V8
VCC_IO3
C72 1uF
C73 1uF
GND
+1V8
RAM
C74 1uF
C75 1uF
C76 1uF
C77 1uF
GND
Note: 3.6V Zener has 3.4V Vz min.
VIOAUX
VCC_IO6
D36 3.6V
C78 1uF
C79 1uF
GND
USB is reverse-current protected, addons needing more than ~2.5W should provide their own power, back-powering the board is allowed. Current draw limited to 0.3A/pin for a total of 1.5A for 5V.
To Test: Adjusted to properly use DQ pins and separate banks but not all data lines are high-speed. Should work but could be slower than 250MHz.
TODO: GND test points near those.
U8E
LFE5U-25F-8BG256x
VCCIO3
PCLKT3_1 L16 CLK0
PR26B L15 DQ0
PR26C M16 DQ4
PCLKC3_0 M15 CLK1
PR29A K13 DQ1
PR29B K12 DQ7
PR29C L13 DQ5
PR29D L12 DQ6
PR32A N16 DQS0
PR32B P15
PR32C L14
PR32D M14 DQ2
PR35A P16 CK0
PR35B R16 CS#0
PR35C M13 DQ3
PR35D N14
PR38A N13 DQ15
PR38B P14 DQ9
PR38C R15 DQ11
PR38D T15 DQ10
PR41A P13 DQ14
PR41B R14 DQ8
PR41C R13 DQ13
PR41D T14 DQ12
PR44A R12 DQS1
PR44B T13 T13
PR44C M12
PR44D N12 N12
PR47A M11 CK1
PR47B N11 N11
PR47C P11 CS#1
PR47D P12 P12
U11
XccelaOPI
U12
XccelaOPI
R260 10k
R261 10k
R262 0
R263 0
R264 0
R265 0
CS#0
CK0
DQS0
CS#1
CK1
DQS1
{DQ[0..15] CS#0 CS#1 CK0 CK1 DQS0 DQS1}
TP7 CS#0
TP8 CK0
TP9 DQS0
TP10 DQ0
TP11 DQ1
TP28 GND
U8F
LFE5U-25F-8BG256x
VCCIO6
J7
PL26A Z14_P
PL26B Z14_N
PL26C Z11_P
PL26D Z11_N
PL29A Z12_P
PL29B Z12_N
PL29C Z15_P
PL29D Z15_N
PL32A Z10_P
PL32B Z10_N
PL32C Z13_P
PL32D Z13_N
PL35A Z6_P
PL35B Z6_N
PL35C Z9_P
PL35D Z9_N
PL38A Z8_P
PL38B Z8_N
PL38C Z3_P
PL38D Z3_N
PL41A Z4_P
PL41B Z4_N
PL41C Z1_P
PL41D Z1_N
PL44A Z2_P
PL44B Z2_N
PL44C Z7_P
PL44D Z7_N
PL47A Z0_P
PL47B Z0_N
PL47C Z5_P
PL47D Z5_N
J6
LVDS
+5V
+3V3
+1V8
+1V1
DETECT
{LVDS}
The LVDS connector J5 just has minimal ESD protection. Use appropriate ESD handling procedures. Plug and unplug only in completely unpowered state.
Suggested mating connector: Hirose DF12NC-60DS-0.5V(51)
H5
Addon Mounting Hole
M2.5 standoff
GND
RAM RESET line left floating, has a weak internal pull-up.
Compatible with:
APS51208N-OBR-BD
APS512XXN-OB9-BG
APS512XXN-OBR-BG
J12
FPGA_TP
C227 12pF
Y4 25M
C228 12pF
C226 1uF
+3V3
+1V8
R266 4.7k
SDA
SCL
OEB
XA
XB
A0
CLK0
CLK1
CLK2
CLK3
CLK_3
VDD
VDDOA
VDDOB
VDDOC
VDDOD
J11
CLK_EXT
D120 3.3V
VCC_CLK_23
C224 1uF
C225 1uF
Y3
Si5351A-B-GM1
Glasgow Debug Tool
In / Fully Automated
Sheet: /FPGA/RAM + LVDS/
File: ram_lvds.kicad_sch
Title: FPGA – RAM and LVDS
Size: A4
Date: 2026-09-22
Rev: D1
KiCad E.D.A. 10.0.6
Id: 5/11

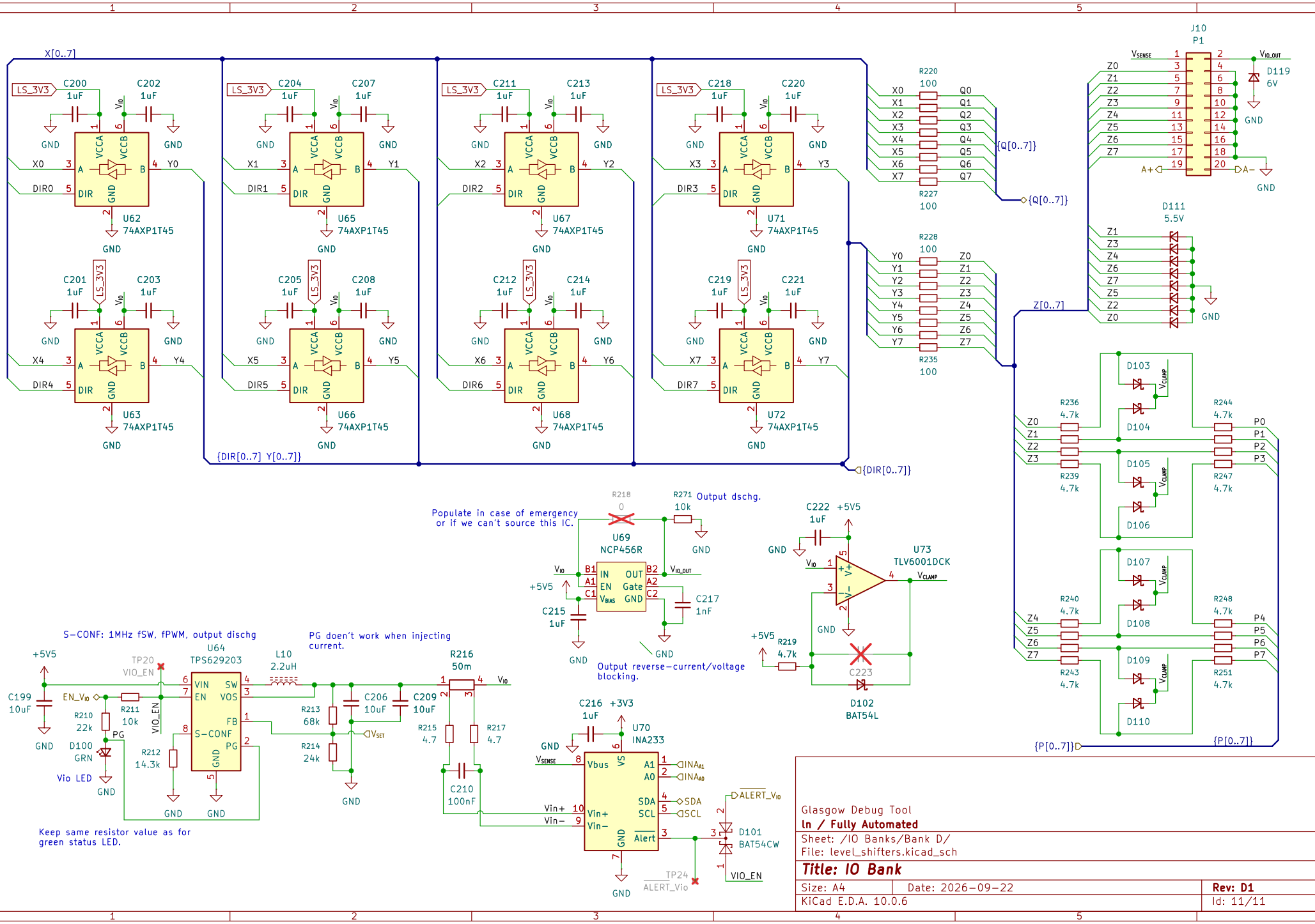

L1
2.2uH
U15
TLV61070ADBV
+5V
C80
10uF
GND
VIN
SW
VOUT
EN
GND
VFB
R65
442k
R66
44.2k
VFB=500mV
C87
360pF
C88
22uF
C91
22uF
C93
22uF
+5V5
Cff= 1 / τ*1kHz*Rtop
Rtop was chosen for a possible combination of E96 resistors and capacitors that match within 1% for the value of Cff.
U20
TPS22917DBV
+3V3
C95
1uF
FPGA_RESETD
VIN
VOUT
CT
ON
GND
QOD
R72
4.7
LS_3V3
left here as an option to change discharge time
VDDA
R73
10k
C100
100nF
U21
DAC43204
C103
1uF
VDD
VREF
GPIO/SDO
SCL/SYNC
SDA/SCLK
A0/SDI
CAP
AGND
OUT0
OUT1
OUT2
OUT3
FB0
FB1
FB2
FB3
VSET_A
VSET_B
VSET_C
VSET_D
R74
10k
SCLD
SDAD
C104
1.5uF
Approx. 1.5uF, fitted 2.2uF
0x48/72/0b1001000
Init, enable ref, Iout ch0, ch1:
1Fh 1Fh F6h
Range +/−50µA:
04h 12h 00h
Output adjust:
40h 1Dh 00h = 5V
40h B7h 00h = 0.9V
Vfb = 0.6V
V_hi = 5.5V
V_lo = 0.9V
V_mid = (V_hi−V_lo)/2 = 5.5V − 0.9V / 2 = 2.3V
R1 = 68k, R2 = 24k
I_margin = (V_out * (1 + MARGIN) − V_fb) / R1 − I_nominal
MARGIN = 100% = 1
I_nominal = V_mid / (R1 + R2) + I_fb
We ignore I_fb because it's 1nA typical (and we keep FBn floating)
I_nominal = 2.3V / (68k + 24k) = 25µA
I_margin = (2.3 * 2 − 0.6) / 68k − I_nominal = 33.8µA
R3 = (V_dac − V_fb) / I_margin = 79.8k
Or TPS62A01
U13
TLV62569PDRL
L2
1uH
C81
4.7uF
R61
453k
R62
100k
C85
22uF
VFB=600mV
CY_RESET
U18
TLV62569PDRL
L4
1uH
C89
4.7uF
R70
100k
R71
120k
C97
22uF
+1V1
U14
TLV62569PDRL
L3
1uH
C82
4.7uF
R63
200k
R64
100k
C86
22uF
+1V8
U17
TLV77525DRL
IN
OUT
EN
GND
C90
1uF
C94
10uF
+2V5
Replace with TLV77525 again for production
net−tie 0−ohm resistors
C17168 / 0402WGF0000TCE
50−mOhm 1/4 W 0402 1% shunt
C851623 / PE0402FRF470R05L
2.2 A limit for power rating
we can disconnect rails, ad−hoc measure currents with a DMM, or we can enamel−wire all the shunts to a dozen INA236s or similar for in−system live power monitoring and optimization of the production power tree
Do we need an LDO? Need to test if this is good enough, or if we need a π−filter or a LDO. Can test on sensacle.
BOIN
L5
10uH
U1SW
D38
30V
+15V
15.0V
TP5
C105
10uF
C106
2.2uF
C109
10uF
C111
10uF
C113
10uF
C115
10uF
U23
SY7208CABC
IN
SW
EN
GND
SNC
FB
0.6V
U1FB
R80
360k
R81
15k
R60
0
R&D
Test w/o U16, should work
Cycle +3.3V on reset to get I2C peri−pherals into clean state.
R69
100k
might be a bit overkill this keeps the RESET low even below 1V
IBBIN
Buck regulator in inverting buck−boost configuration.
Vin min = 4 V
U22
LMR54406
C107
100nF/50V
L6
10uH
U2SW
C98
10uF
C101
2.2uF
C102
2.2uF
EN
BS
SW
FB
R76
100k
R75
100k
R77
100k
Q1
AO3401
Q2
AO3400
INVBUBO_LOCALGND
0.8V
U2FB
R78
390k
C108
R79
22k
C110
C112
C114
D39
24V
TP6
−14.99V
−15V
+5V − −15V = 20V across Vgs would kill the nMOS (±12V max)
I might add a zener here but the divider shall do
C84
100nF
U16
APX811−40UG−7
MR
VCC
RESET
GND
R59
100k
PGVBUS
PGVBUS needs bus power to start up and does not set PG high even if the 5V rail is stable but bus voltage is not present.
push−pull output
Turn on at VBUS >4.0V
SW1
RST
C83
100nF
Keep CY_RESET active for >5 msec after 3.3V switches on
D37
BAT54AW
??
R67
100k
R68
100k
C92
100nF
A reset of U16 switches off the Vio regulators due to pulldowns. The RST button is designed to be usable as emergency off button.
FB2
120R@100MHz
C96
1uF
U19
TPS7A2033
VIN
VOUT
EN
GND
C99
10uF
VDDA
Glasgow Debug Tool
In / Fully Automated
Sheet: /Power/
File: power.kicad_sch
Title: Power Supplies
Size: A4
Date: 2026−09−22
Rev: D1
KiCad E.D.A. 10.0.6
Id: 6/11


IO Bank Decoupling
+3V3
C116 1uF
C118 1uF
C117 1uF
C119 1uF
GND
U8D
LFE5U-25F-8BG256x
VCCIO2
U8G
LFE5U-25F-8BG256x
VCCIO7
A{DIR[0..7] Q[0..7]}
B{DIR[0..7] Q[0..7]}
C{DIR[0..7] Q[0..7]}
D{DIR[0..7] Q[0..7]}
0x20/32/0b0100000
0x21/33/0b0100001
C120+3V3+5V5C122
C121+3V3+5V5C123
R82 10k
R83 10k
R258 0
R268 0
ALERT
SCL
SDA
INT
RESET
ADDR
VDD_I2C
VDD_P
VSS
U24
U25
TCA6416APW
Swapped NXP with TI part for availability.
Potential alternative: NCA6416, PCA6416A
INA A: 1001100
Bank A
INA B: 1001101
Bank B
INA C: 1001110
Bank C
INA D: 1001111
Bank D
File: level_shifters.kicad_sch
{ADC}
IO&DIR should share a PIO pair if possible
in any case, no PIO pair may be shared between different
bank pins
Glasgow Debug Tool
In / Fully Automated
Sheet: /IO Banks/
File: io_buffer.kicad_sch
Title: FPGA – IO Banks & Port Expander
Size: A4
Date: 2026-09-22
Rev: D1
KiCad E.D.A. 10.0.6
Id: 7/11


X[0..7]
LS_3V3
C125 1uF
C127 1uF
C129 1uF
C132 1uF
C136 1uF
C138 1uF
C143 1uF
C145 1uF
C126 1uF
C128 1uF
C130 1uF
C133 1uF
C137 1uF
C139 1uF
C144 1uF
C146 1uF
U26 74AXP1T45
U27 74AXP1T45
U29 74AXP1T45
U30 74AXP1T45
U31 74AXP1T45
U32 74AXP1T45
U35 74AXP1T45
U36 74AXP1T45
GND
{DIR[0..7] Y[0..7]}
{DIR[0..7]}
R94 100
R101 100
R102 100
R109 100
{Q[0..7]}
Z[0..7]
J7
P1
D59 6V
D51 5.5V
D43
D44
D45
D46
D47
D48
D49
D50
R110 4.7k
R113 4.7k
R114 4.7k
R117 4.7k
R118 4.7k
R121 4.7k
R122 4.7k
R125 4.7k
{P[0..7]}
Populate in case of emergency or if we can't source this IC.
R92 0
R259 10k
Output dschg.
U33 NCP456R
C140 1uF
C142 1nF
Output reverse-current/voltage blocking.
C147 1uF
U37 TLV6001DCK
R93 4.7k
C148
D42 BAT54L
S-CONF: 1MHz fSW, fPWM, output dischg
PG doen't work when injecting current.
U28 TPS629203
L7 2.2uH
C124 10uF
R84 22k
R85 10k
R86 14.3k
R87 68k
R88 24k
C131 10uF
C134 10uF
R90 50m
R89 4.7
R91 4.7
C135 100nF
D40 GRN
Vio LED
TP17 VIO_EN
TP21 ALERT_Vio
Keep same resistor value as for green status LED.
C141 1uF
U34 INA233
D41 BAT54CW
Glasgow Debug Tool
In / Fully Automated
Sheet: /IO Banks/Bank A/
File: level_shifters.kicad_sch
Title: IO Bank
Size: A4
Date: 2026-09-22
Rev: D1
KiCad E.D.A. 10.0.6
Id: 8/11


X[0..7]
LS_3V3
C150 1uF
C152 1uF
C154 1uF
C157 1uF
C161 1uF
C163 1uF
C168 1uF
C170 1uF
C151 1uF
C153 1uF
C155 1uF
C158 1uF
C162 1uF
C164 1uF
C169 1uF
C171 1uF
VIO
VCCA
VCCB
GND
A
B
DIR
U38 74AXP1T45
U39 74AXP1T45
U41 74AXP1T45
U42 74AXP1T45
U43 74AXP1T45
U44 74AXP1T45
U47 74AXP1T45
U48 74AXP1T45
X0
X1
X2
X3
X4
X5
X6
X7
Y0
Y1
Y2
Y3
Y4
Y5
Y6
Y7
DIR0
DIR1
DIR2
DIR3
DIR4
DIR5
DIR6
DIR7
{DIR[0..7] Y[0..7]}
{DIR[0..7]}
R136 100
R143 100
R144 100
R151 100
{Q[0..7]}
Z[0..7]
J8
P1
VSENSE
VIO_OUT
D79 6V
D71 5.5V
A+
A-
D63
D64
D65
D66
D67
D68
D69
D70
VCLAMP
R152 4.7k
R155 4.7k
R156 4.7k
R159 4.7k
R160 4.7k
R163 4.7k
R164 4.7k
R167 4.7k
P0
P1
P2
P3
P4
P5
P6
P7
{P[0..7]}
Populate in case of emergency
or if we can't source this IC.
R134 0
R269 10k
Output dschg.
U45
NCP456R
IN
OUT
EN
Gate
VBIAS
C165 1uF
C167 1nF
Output reverse-current/voltage
blocking.
C172 1uF
+5V5
U49
TLV6001DCK
R135 4.7k
C173
D62
BAT54L
S-CONF: 1MHz fSW, fPWM, output dischg
PG doen't work when injecting
current.
U40
TPS629203
VIN
SW
EN
VOS
FB
S-CONF
PG
L8 2.2uH
R132 50m
C149 10uF
C156 10uF
C159 10uF
R126 22k
R127 10k
R128 14.3k
R129 68k
R130 24k
R131 4.7
R133 4.7
C160 100nF
TP18
VIO_EN
EN_VIO
PG
D60 GRN
Vio LED
VSET
C166 1uF
+3V3
U46
INA233
Vbus
VS
A1
A0
SDA
SCL
Alert
Vin+
Vin-
INA_A1
INA_A0
SDA
SCL
ALERT_VIO
D61 BAT54CW
TP22
ALERT_Vio
Keep same resistor value as for
green status LED.
Glasgow Debug Tool
In / Fully Automated
Sheet: /IO Banks/Bank B/
File: level_shifters.kicad_sch
Title: IO Bank
Size: A4
Date: 2026-09-22
Rev: D1
KiCad E.D.A. 10.0.6
Id: 9/11


X[0..7]
LS_3V3
C175 1uF
C177 1uF
U50 74AXP1T45
C176 1uF
C178 1uF
U51 74AXP1T45
C179 1uF
C182 1uF
U53 74AXP1T45
C180 1uF
C183 1uF
U54 74AXP1T45
C186 1uF
C188 1uF
U55 74AXP1T45
C187 1uF
C189 1uF
U56 74AXP1T45
C193 1uF
C195 1uF
U59 74AXP1T45
C194 1uF
C196 1uF
U60 74AXP1T45
GND
VCCA
VCCB
DIR
{DIR[0..7] Y[0..7]}
R178 100
R185 100
{Q[0..7]}
R186 100
R193 100
Z[0..7]
{DIR[0..7]}
J9
P1
D99 6V
D91 5.5V
D83
D84
D85
D86
D87
D88
D89
D90
R194 4.7k
R197 4.7k
R198 4.7k
R201 4.7k
R202 4.7k
R205 4.7k
R206 4.7k
R209 4.7k
{P[0..7]}
Populate in case of emergency or if we can't source this IC.
R176 0
R270 10k Output dschg.
U57 NCP456R
C190 1uF
C192 1nF
Output reverse-current/voltage blocking.
C197 1uF
U61 TLV6001DCK
R177 4.7k
C198
D82 BAT54L
S-CONF: 1MHz fSW, fPWM, output dischg
PG doen't work when injecting current.
U52 TPS629203
L9 2.2uH
R174 50m
C174 10uF
R168 22k
R169 10k
R170 14.3k
D80 GRN
Vio LED
R171 68k
R172 24k
C181 10uF
C184 10uF
R173 4.7
R175 4.7
C185 100nF
Keep same resistor value as for green status LED.
C191 1uF
U58 INA233
D81 BAT54CW
TP23 ALERT_Vio
Glasgow Debug Tool
In / Fully Automated
Sheet: /IO Banks/Bank C/
File: level_shifters.kicad_sch
Title: IO Bank
Size: A4
Date: 2026-09-22
Rev: D1
KiCad E.D.A. 10.0.6
Id: 10/11


X[0..7]
LS_3V3
C200 1uF
C202 1uF
C201 1uF
C203 1uF
C204 1uF
C207 1uF
C205 1uF
C208 1uF
C211 1uF
C213 1uF
C212 1uF
C214 1uF
C218 1uF
C220 1uF
C219 1uF
C221 1uF
U62 74AXP1T45
U63 74AXP1T45
U65 74AXP1T45
U66 74AXP1T45
U67 74AXP1T45
U68 74AXP1T45
U71 74AXP1T45
U72 74AXP1T45
GND
VCCA
VCCB
DIR
{DIR[0..7] Y[0..7]}
R220 100
R227 100
R228 100
R235 100
{Q[0..7]}
{DIR[0..7]}
J10
P1
D119 6V
D111 5.5V
Z[0..7]
R236 4.7k
R239 4.7k
R244 4.7k
R247 4.7k
R240 4.7k
R243 4.7k
R248 4.7k
R251 4.7k
D103
D104
D105
D106
D107
D108
D109
D110
{P[0..7]}
Populate in case of emergency
or if we can't source this IC.
R218 0
R271 10k Output dschg.
U69 NCP456R
C215 1uF
C217 1nF
Output reverse-current/voltage
blocking.
C222 1uF
U73 TLV6001DCK
R219 4.7k
C223
D102 BAT54L
S-CONF: 1MHz fSW, fPWM, output dischg
PG doen't work when injecting
current.
U64 TPS629203
L10 2.2uH
R216 50m
C199 10uF
R210 22k
R211 10k
R212 14.3k
R213 68k
R214 24k
C206 10uF
C209 10uF
R215 4.7
R217 4.7
C210 100nF
D100 GRN
Vio LED
TP20 VIO_EN
Keep same resistor value as for
green status LED.
C216 1uF
U70 INA233
TP24 ALERT_Vio
D101 BAT54CW
Glasgow Debug Tool
In / Fully Automated
Sheet: /IO Banks/Bank D/
File: level_shifters.kicad_sch
Title: IO Bank
Size: A4
Date: 2026-09-22
Rev: D1
KiCad E.D.A. 10.0.6
Id: 11/11